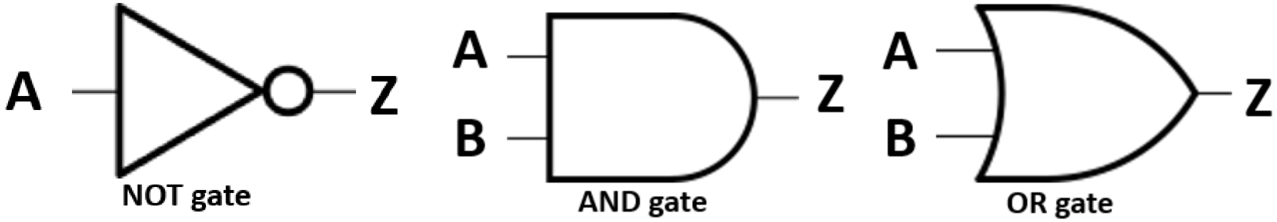
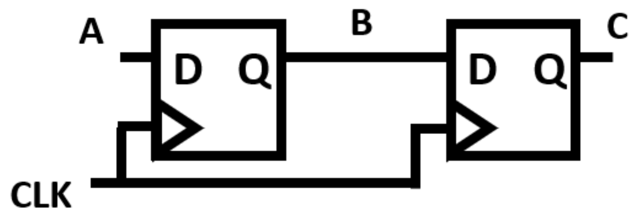
EEE 143: Switching Theory and Digital Logic Design

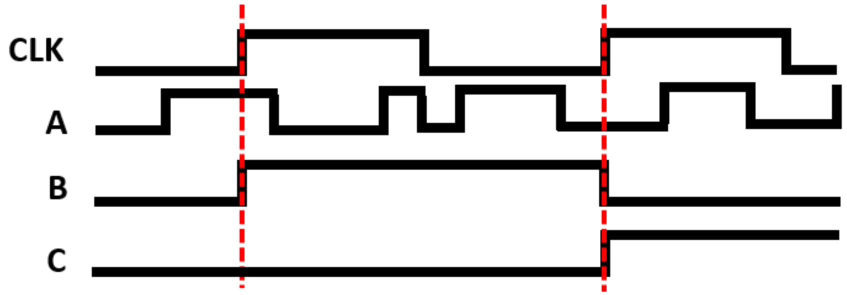
Lecture 0: Course Introduction

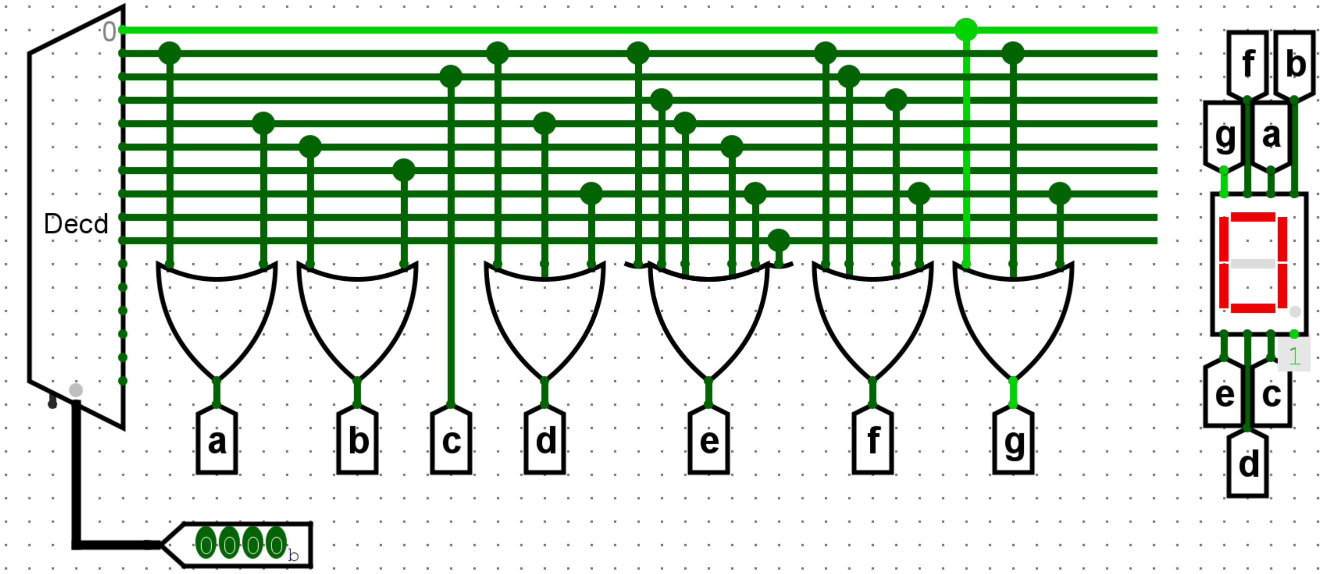
What is this class all about?

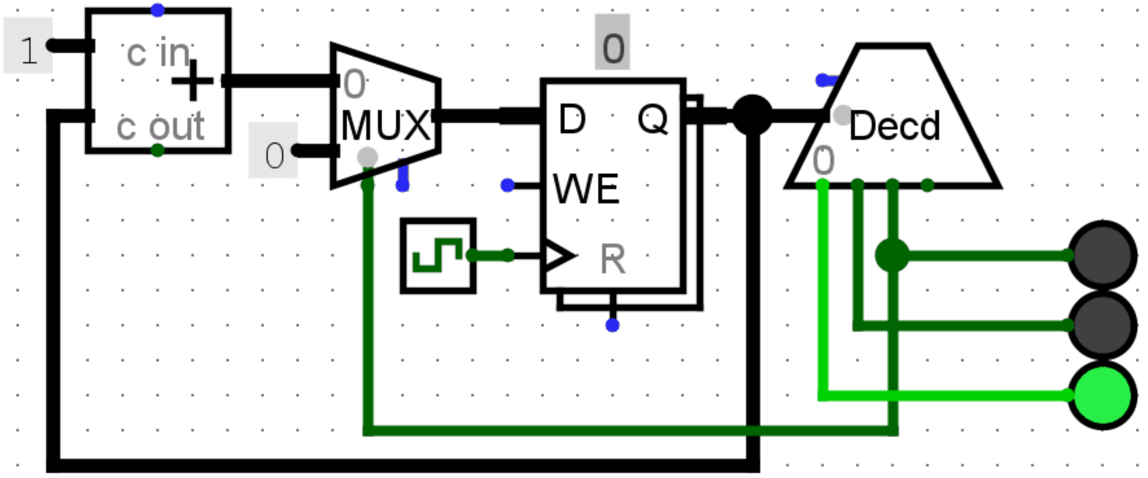
- Remember EEE 113 (Computing)?











Introduction (from EEE 113)

- There are several abstraction layers in a computer system
  - EEE 111 is on the highest level (high-level programming using Python, C, etc)
  - EEE 113 briefly touched the lower levels of the hierarchy... **in 4 weeks**!
    - How a computer *really* does what it does.

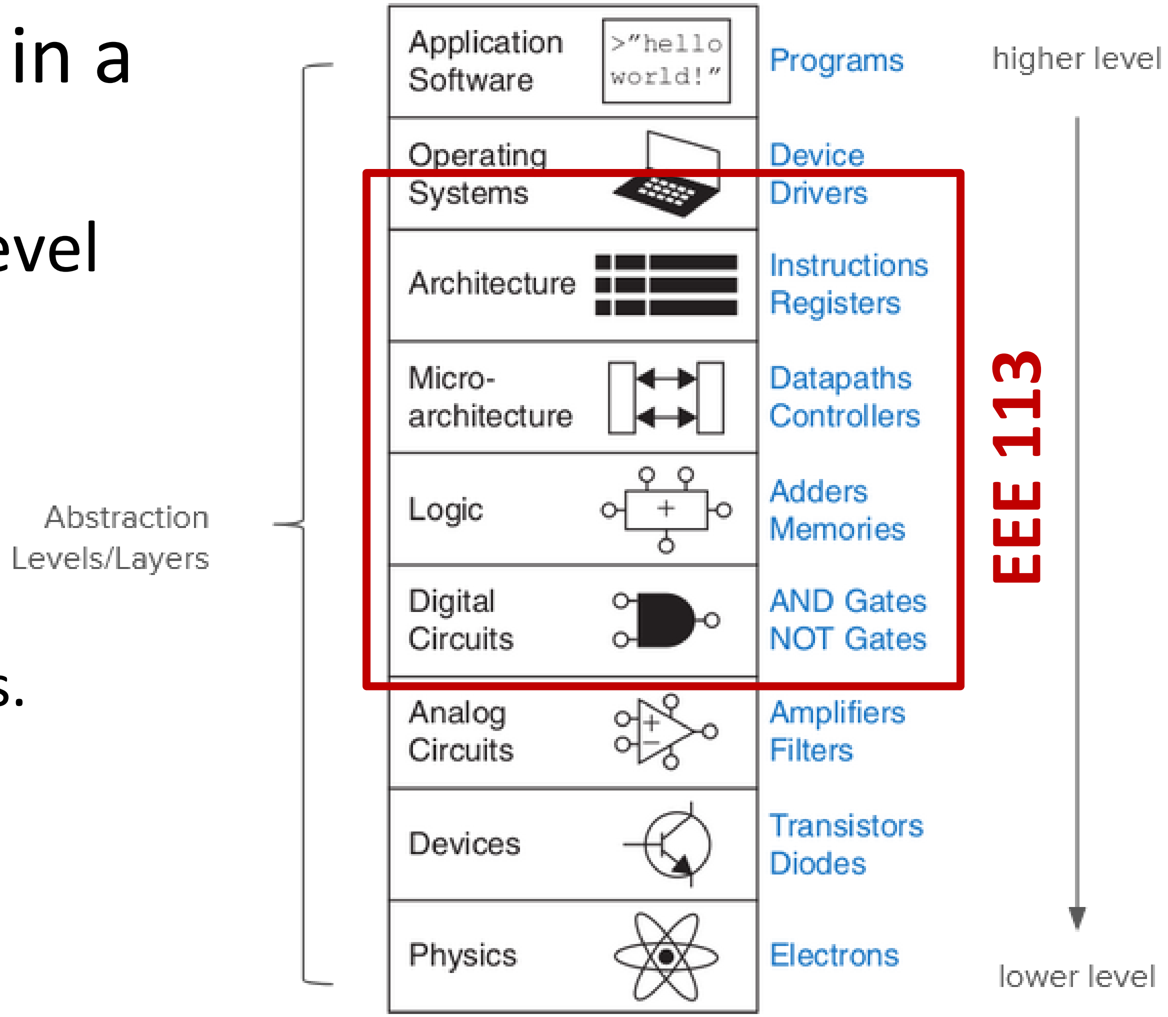


Image credit: *Digital Design and Computer Architecture* (2nd ed.) by Harris, H., Harris, S. (2013).

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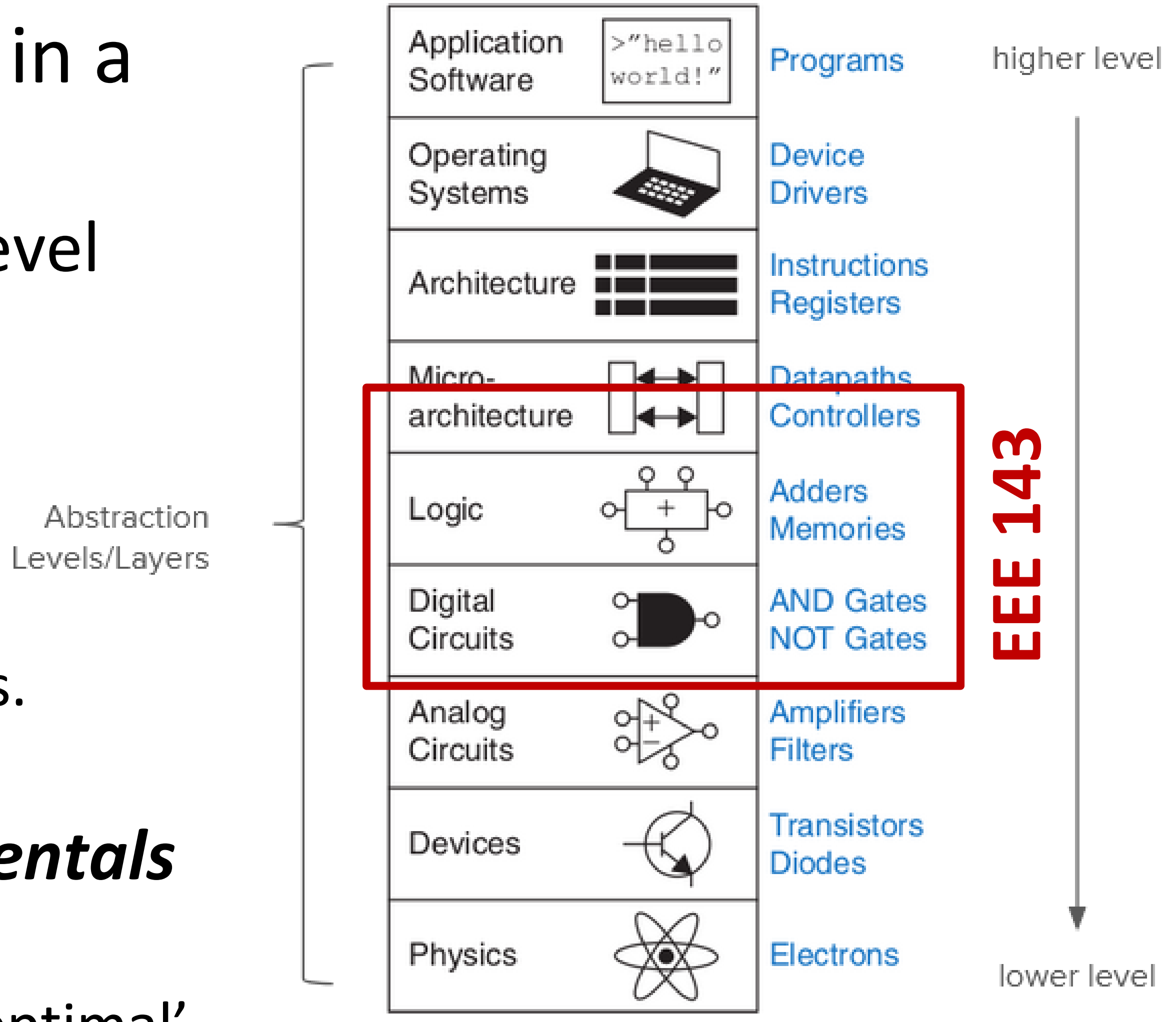

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What is this class all about?

- Logic gates, Boolean algebra, combinational logic blocks
- Sequential circuits, analysis and synthesis techniques, special topics

- Basically, **EEE 113 (Computing)** but we'll go <u>**in-depth**</u>!

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    - Number systems and representations (*how to represent fractional numbers?*)

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    - Boolean manipulation techniques *(how to efficiently minimize expressions?)*

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    - Circuit non-idealities *(how do non-idealities influence logic gate delays?)*

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    - Circuit non-idealities
    - Analyzing and synthesizing sequential circuits *(how to implement sequential circuits?)*

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    - Analyzing and synthesizing sequential circuits
    - Sequential circuit performance metrics *(how do non-idealities influence the clock?)*

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  - Learn more about:
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    - Circuit non-idealities
    - Analyzing and synthesizing sequential circuits
    - Sequential circuit performance metrics
    - Alternative sequential circuit design approaches leading toward processor design (*how can we apply all these techniques to implement higher-level systems?*)

Course coverage

Topic	No. of hours
Introduction; Number systems and representations	3
Logic gates; Boolean algebra; Karnaugh maps	9
Combinational logic blocks; logic non-idealities	7.5
Exam 1	
Sequential circuits: Analysis, Minimization, Synthesis	10.5
Sequential circuit non-idealities, design considerations	3
Special topics: Algorithmic state machines; HDL; Processors	6
Exam 2	

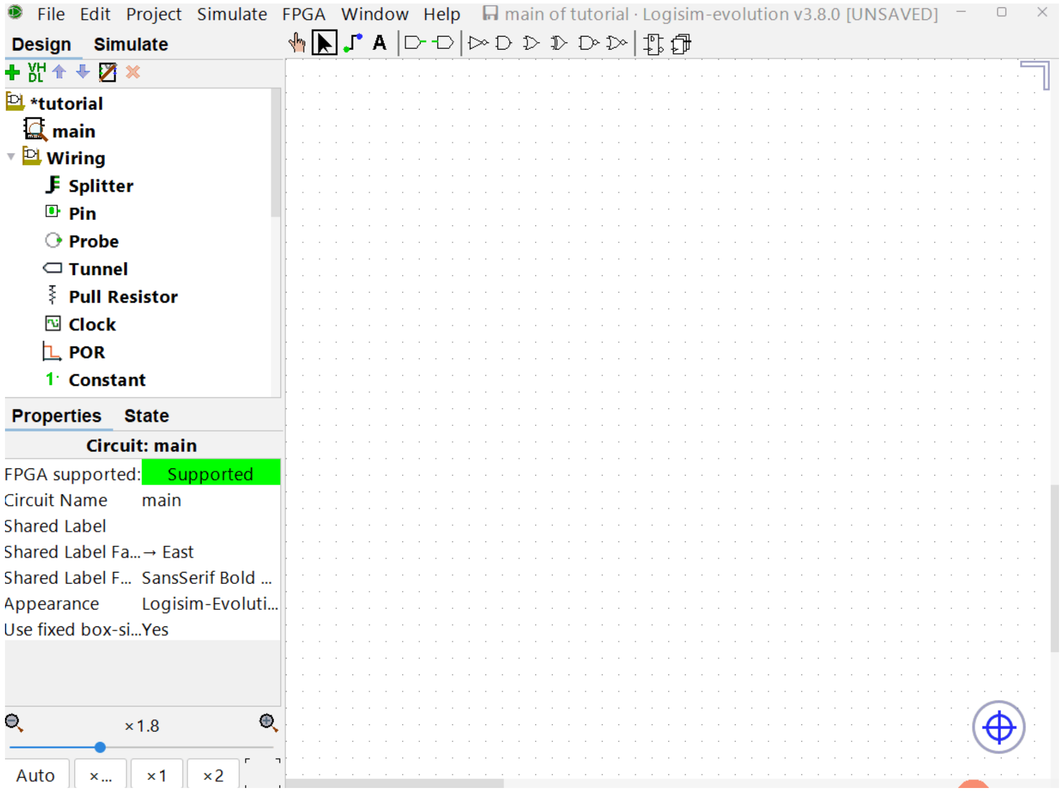
Course requirements

- **60% Long Exams (2)**
  - March 15 (Saturday): 1 - 4 pm*
  - May 24 (Saturday): 1 - 4 pm*
- **30% Quizzes/Homework (weekly)**
  - UVLe online quizzes, file submissions
  - Logic simulation exercises
- **10% In-class discussions/activities**

*Provisions for early/late exam in case of conflicts

Logic simulation software

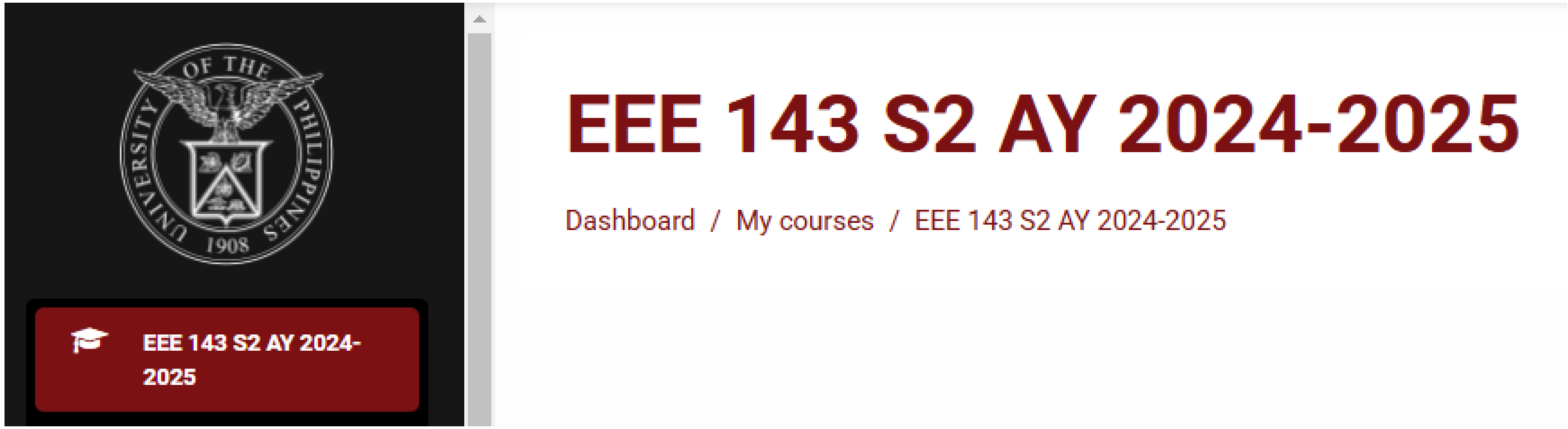
- We'll be using Logisim Evolution for logic simulations
  - (https://github.com/logisim-evolution/logisim-evolution)
  - We'll have *a lot* of exercises using this software
  - **Visualize what we are learning about!**



Installation instructions will be provided in UVLe!

Class page and resource

- We'll be using UVLe (https://uvle.upd.edu.ph)
  - Lecture slides and other resource materials
  - Online quizzes and homework submission bins
  - Class announcements



We will be enrolling all of you by the end of this week!

Online forum

- We’ll use **Piazza** and **UVLe forums** for Q&A
  - Don’t be shy about asking questions/seeking clarifications!
    - Questions and answers can be seen by the entire class. Equitable and better visibility!
    - You might have a question that your classmates also wonder about...
    - Someone might have a question that never even crossed your mind...
  - You can ask anonymously (the instructors can still see who it is though)
  - We’ll try to answer questions but your classmates can also help answer.
  - Participate and contribute to the learning community!

EEE 143: Switching Theory and Digital Logic Design

Lecture 0: Course Introduction
(*Reminiscing EEE 113: Computing*)

Computer systems

- *Computers* are machines that were invented to help make the *"thinking work" easier and faster*

Solving equations

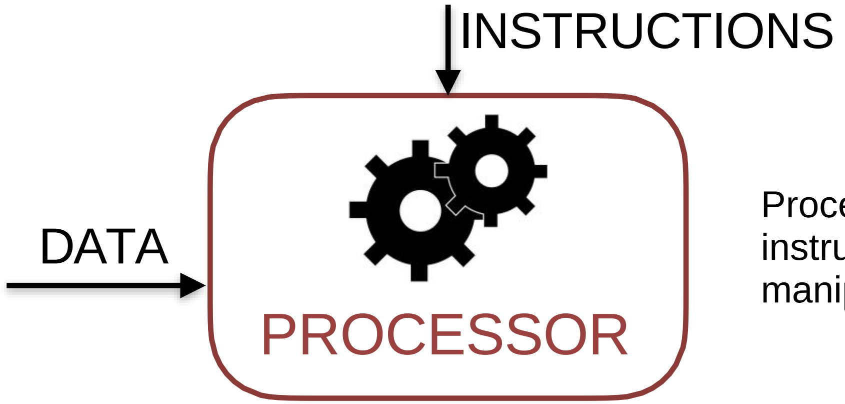
Tracking information

Analyzing information

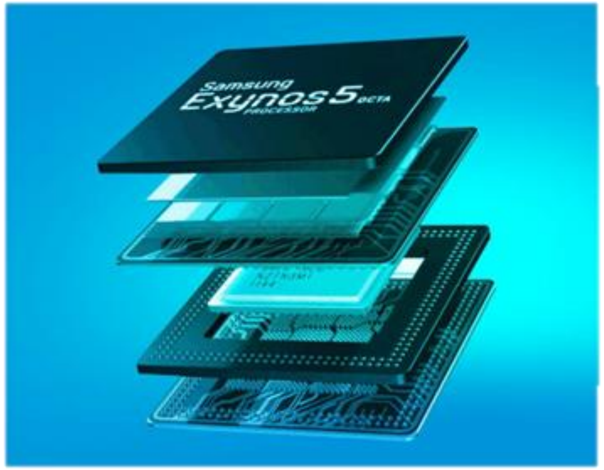
A computer system

PROCESSOR

- At the heart of a computer system is the **processor**
  - Contains several functional units for performing operations on data
  - Has internal memory called registers to contain operands



Processors need instructions to properly manipulate data.

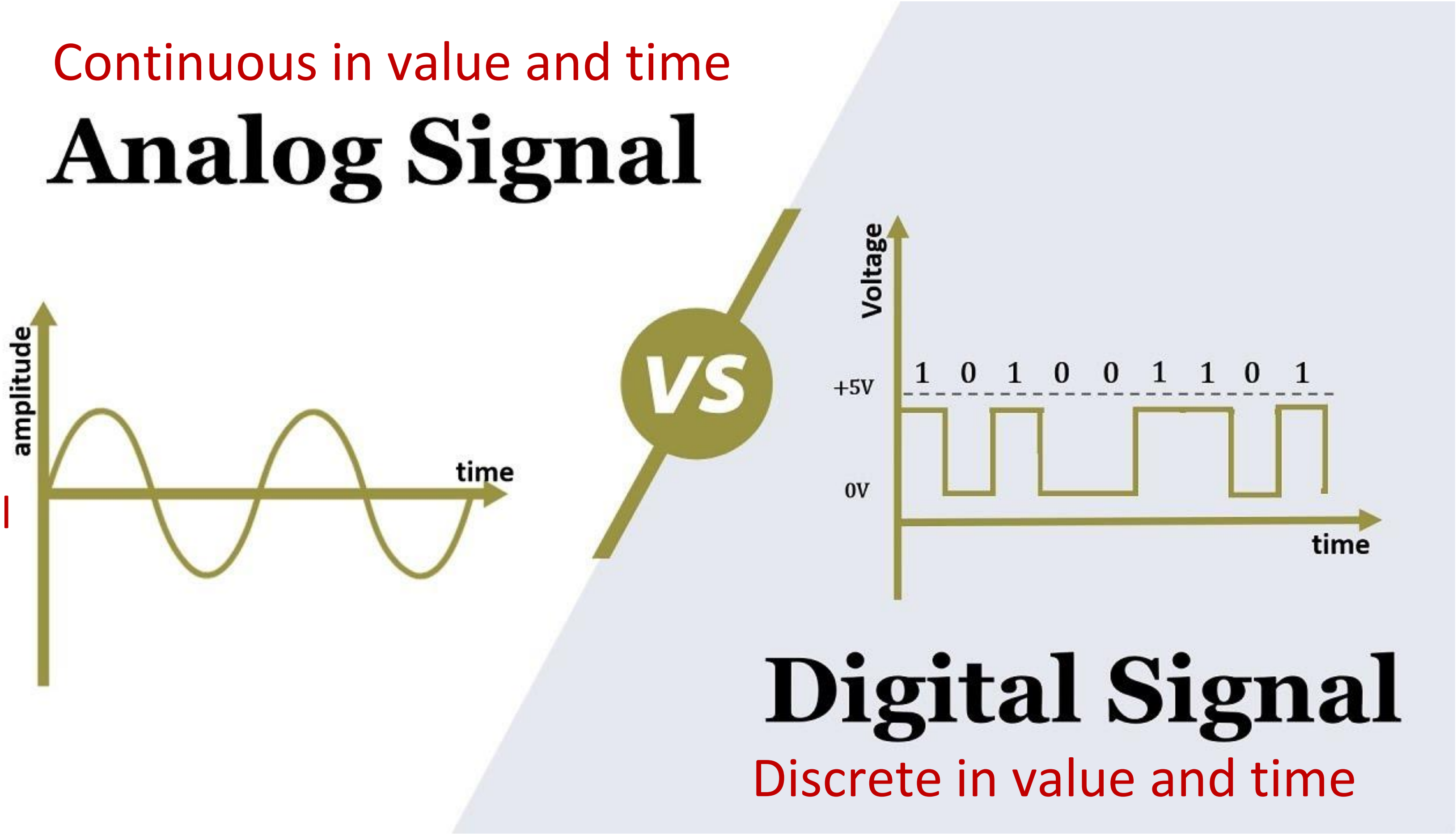


EEE 113: Computing has shown how processors really just *manipulate 0s and 1s*

Analog vs Digital signals

- Information can be transferred either through **analog** or **digital** means

Continuous in value and time



Discrete in value and time

Responses from changes in physical phenomena:
- Temperature
- Voltage/current
- etc.

More resilient to noise compared to analog signals

Computer systems utilize digital signals!

Analog information can be digitized: Color

- Color information can be displayed by forming a pixel made of red, green, and blue (**RGB**) light-emitting circuit elements
  - Each color can vary in intensity, typically ranging from [0,255].

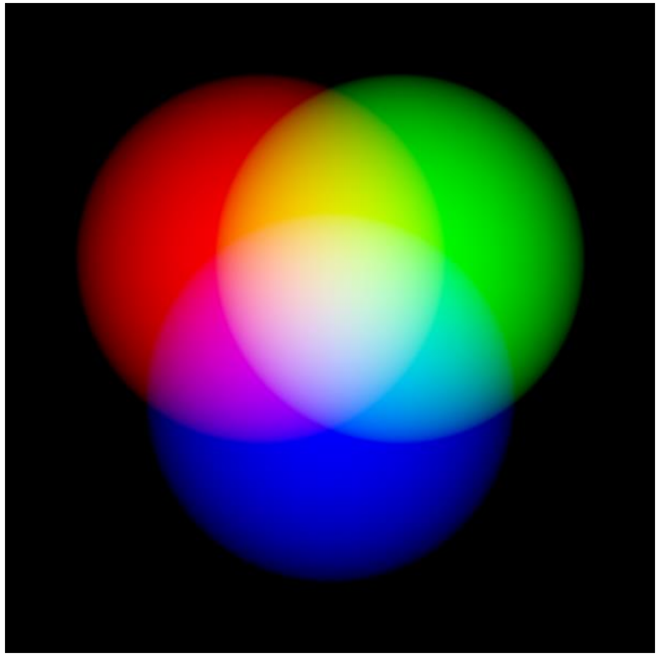
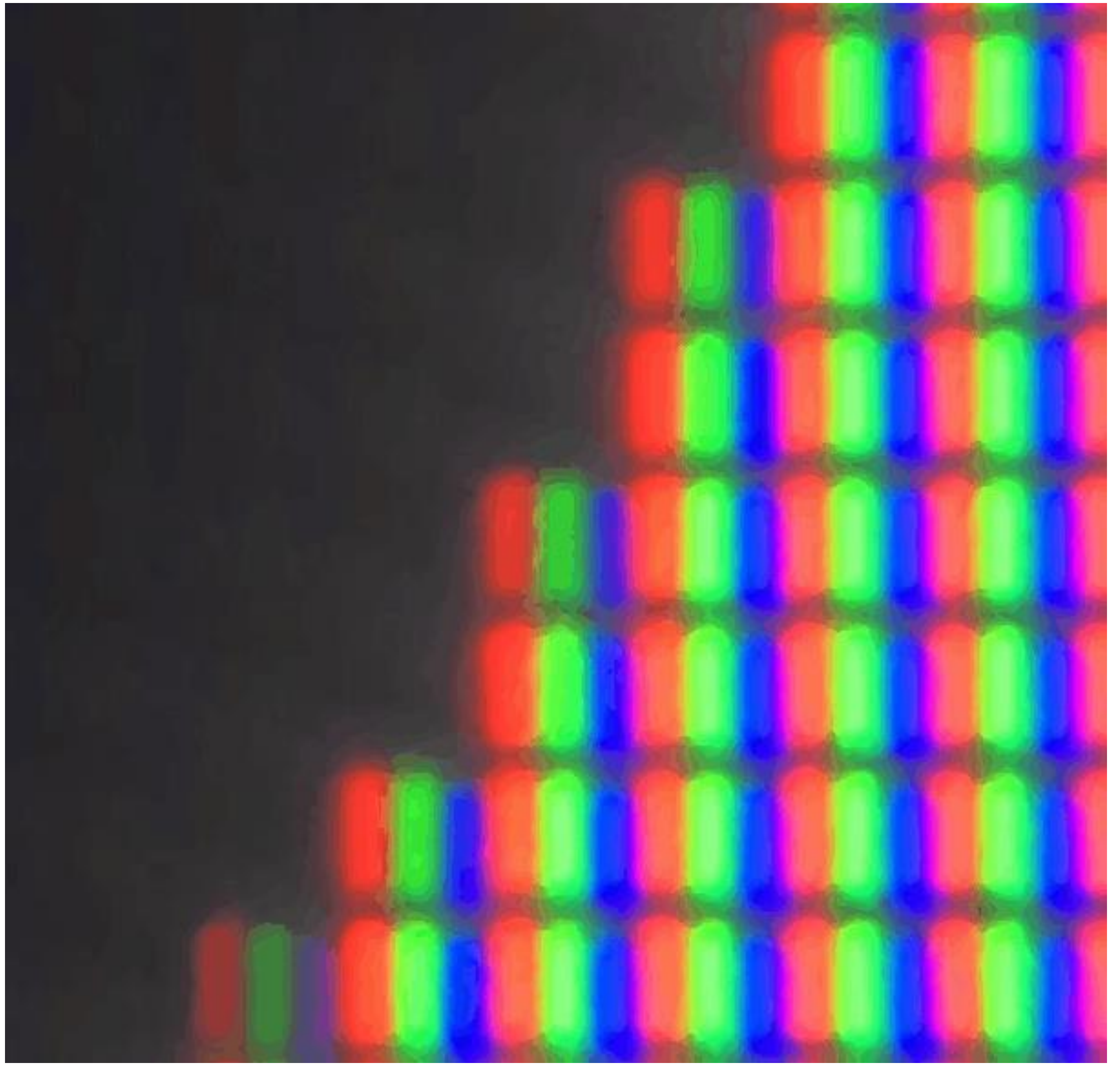


Image credit: commons.wikimedia.org



Video from: https://www.youtube.com/shorts/T9ryMOrwVZQ

RGB display demo ☺

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	25	20	22	19	22	24	22	83	15	20	80	80	86	70	
2	28	28	28	29	30	32	32	100	26	29	97	100	131	117	
3	19	13	18	18	19	17	21	126	32	44	123	151	198	189	1
4	27	17	29	14	22	20	18	68	14	72	49	104	94	77	
5	30	28	40	24	30	34	27	92	26	96	67	151	150	122	
6	19	11	26	15	19	33	22	102	42	132	105	219	227	181	1
7	18	19	34	12	15	17	44	23	13	66	8	53	35	77	
8	24	30	44	19	28	33	61	33	25	83	28	76	71	105	1
9	12	16	33	12	34	33	71	33	47	103	52	118	119	152	1
10	30	18	46	28	45	87	15	17	84	29	53	23	83	56	
11	38	29	66	44	72	112	30	27	112	61	89	50	108	90	1
12	23	21	77	44	89	143	53	26	160	110	141	61	146	151	2
13	22	23	34	28	38	38	36	52	97	46	82	46	106	106	
14	33	34	54	37	56	50	49	69	137	84	118	64	136	152	1
15	27	26	65	32	42	40	40	79	186	147	168	86	188	227	1
16	13	36	30	63	33	21	32	82	73	76	80	79	60	96	
17	25	40	34	69	39	24	43	96	91	126	122	111	98	152	1
18	13	25	17	57	25	17	39	60	95	199	188	170	161	225	1
19	47	43	25	25	18	71	78	69	55	101	98	97	77	93	
20	58	49	34	33	21	74	86	84	78	150	148	115	99	142	1
21	54	[illegible]	[illegible]	[illegible]	[illegible]	65	61	51	84	217	207	151	123	209	2

Image

Ready Accessibility: Good to go

400%

https://think-maths.co.uk/spreadsheet/

Analog information can be digitized: Sound

- Sound can be digitized through *analog-to-digital converters* (ADCs) and is converted back to sound waves through *digital-to-analog converters* (DACs)

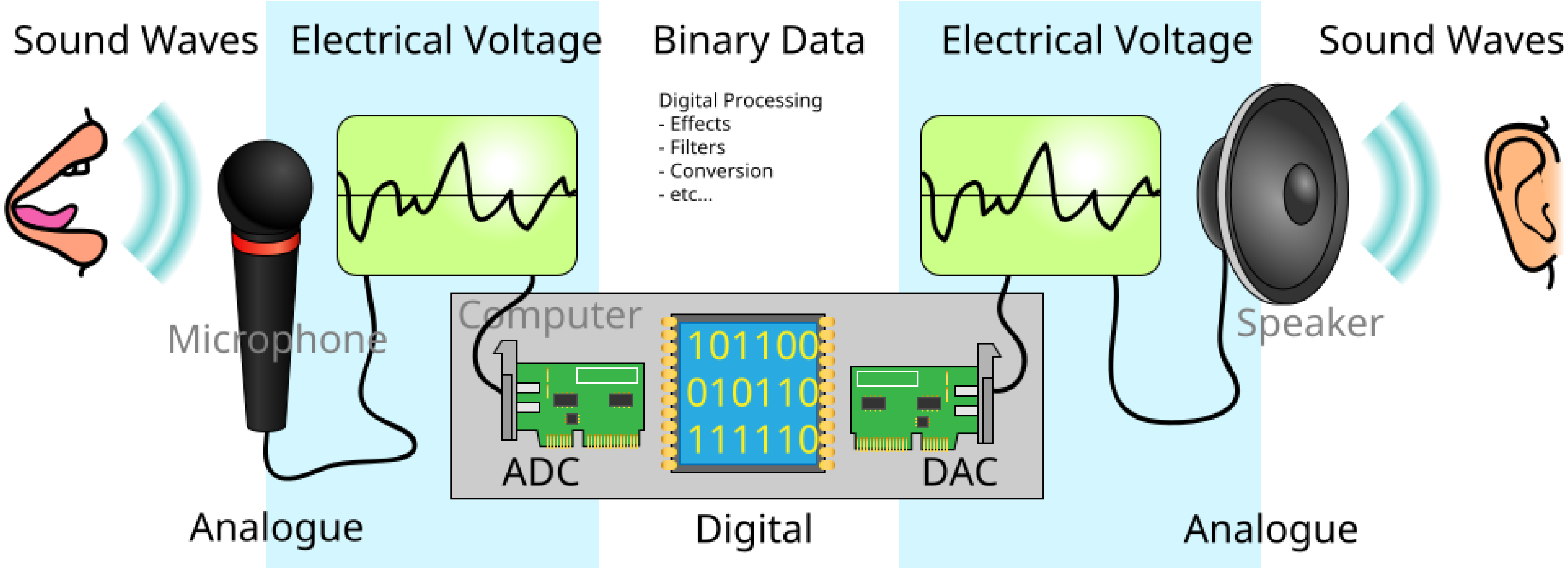
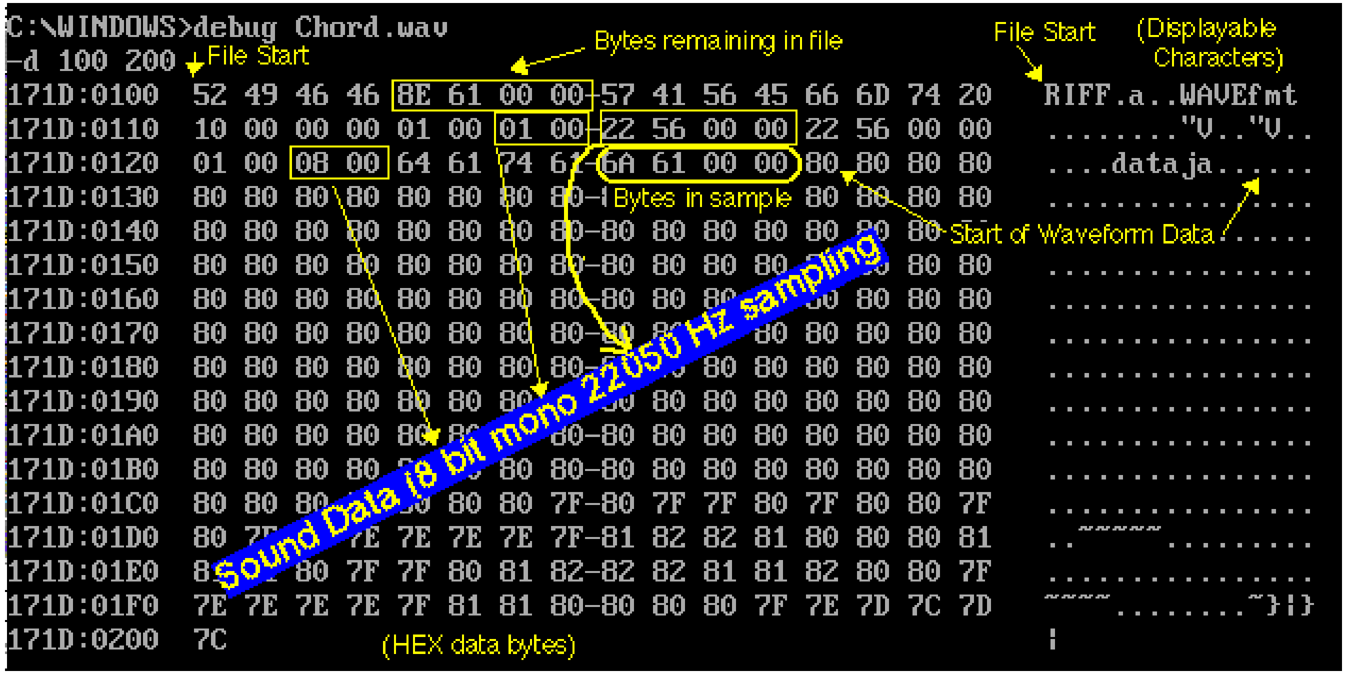


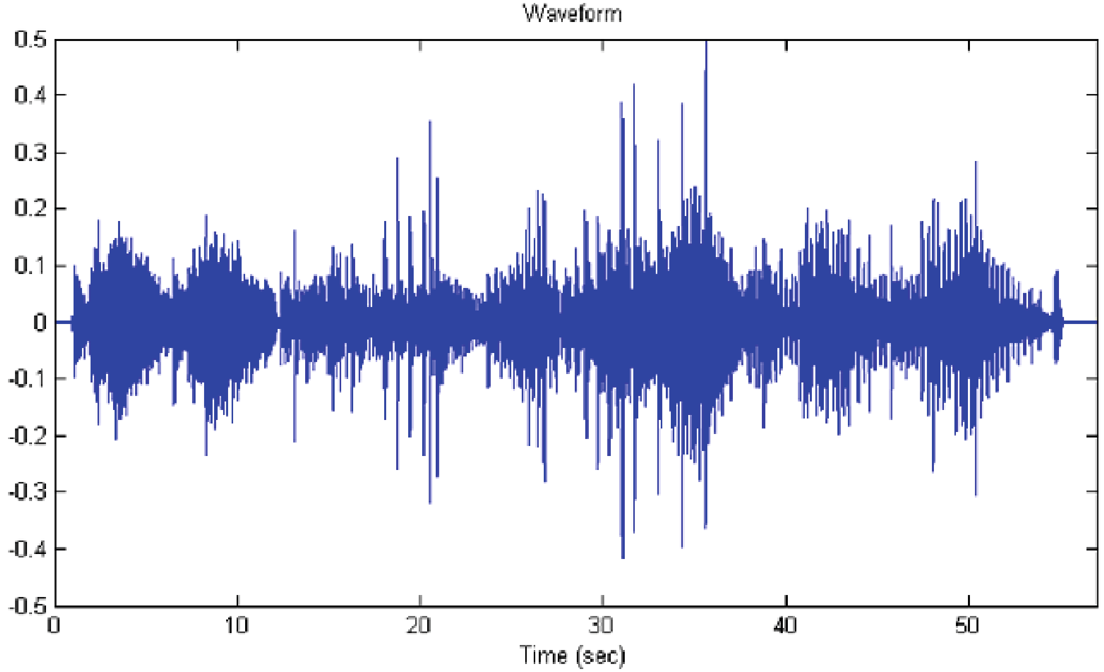
Image credit: wikibooks.org

Digitizing sound

- In a computer, sound files are just binary representing the waveform!
  - Some file formats are compressed versions, but they're still waveform data in the end.



WAVE (.wav) data format including the file headers



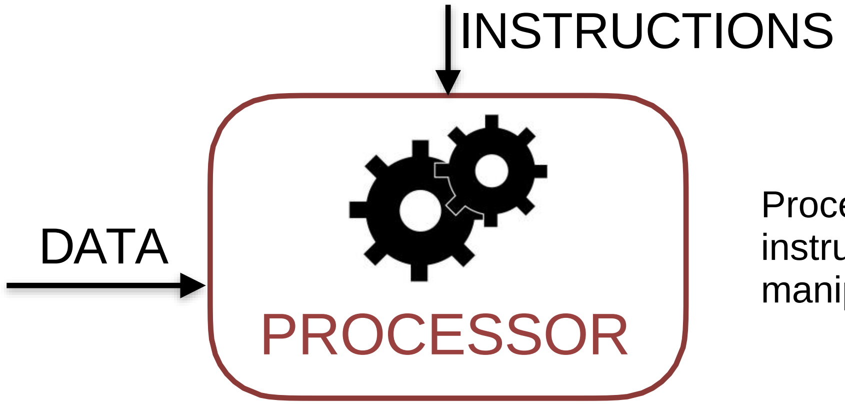
Plotting the .wav file after removing the file headers

Image credit: https://www.jensign.com/www/mmedia/waveinfo.html

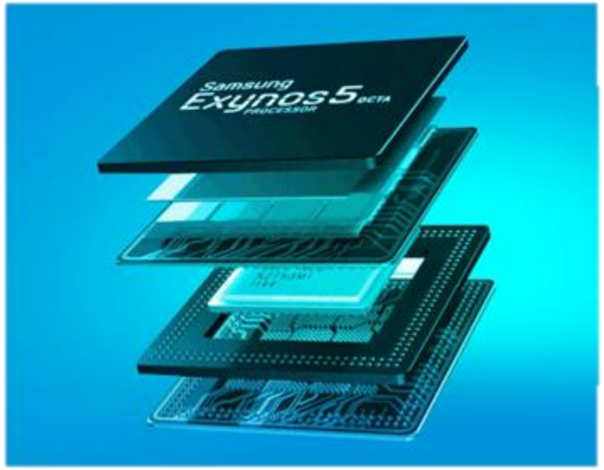
A computer system

PROCESSOR

EEE 113: Computing has shown how processors really just manipulate 0s and 1s



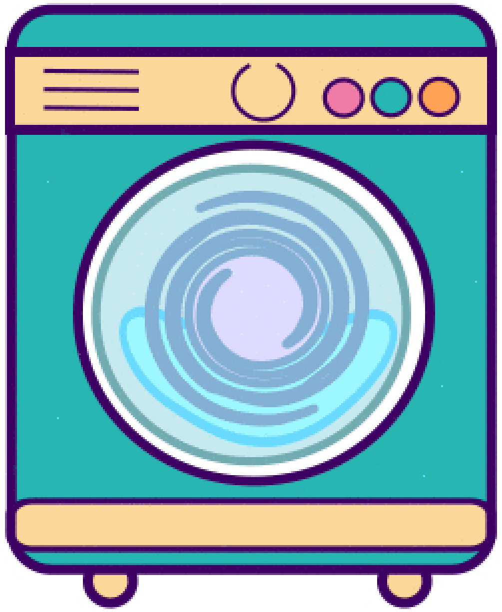
Processors need instructions to properly manipulate data.

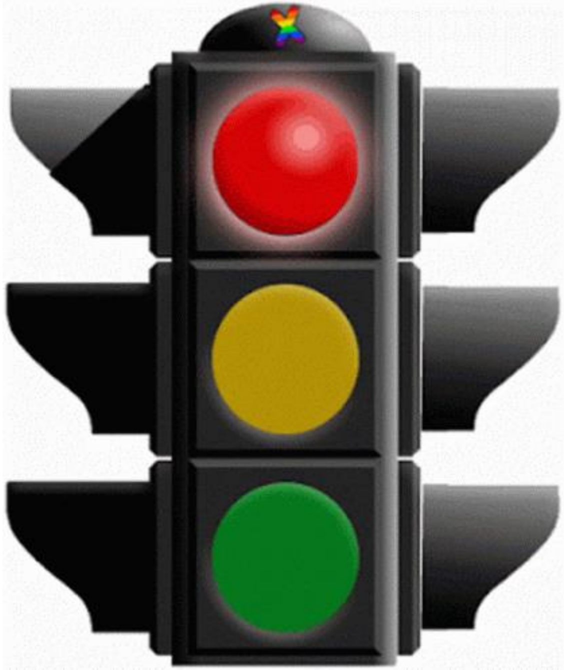


Computers are *digital systems*

Digital systems

- The basic building block of a digital system is a switch
  - Two states: **OFF** or **ON**... **LOW** or **HIGH**... **0** or **1** (binary digits)
  - Digital systems manipulate digitized data using <u>logic</u>





Digital systems are all around us!

GIFs from tenor.com

Digital systems

- EEE 143 focuses more on the ***fundamentals*** of digital logic design
  - Design techniques that make a system 'optimal'

- EEE 143 answers the question:

How do we optimally create
a *network of switches*
that will allow us
to *manipulate* digitized data?

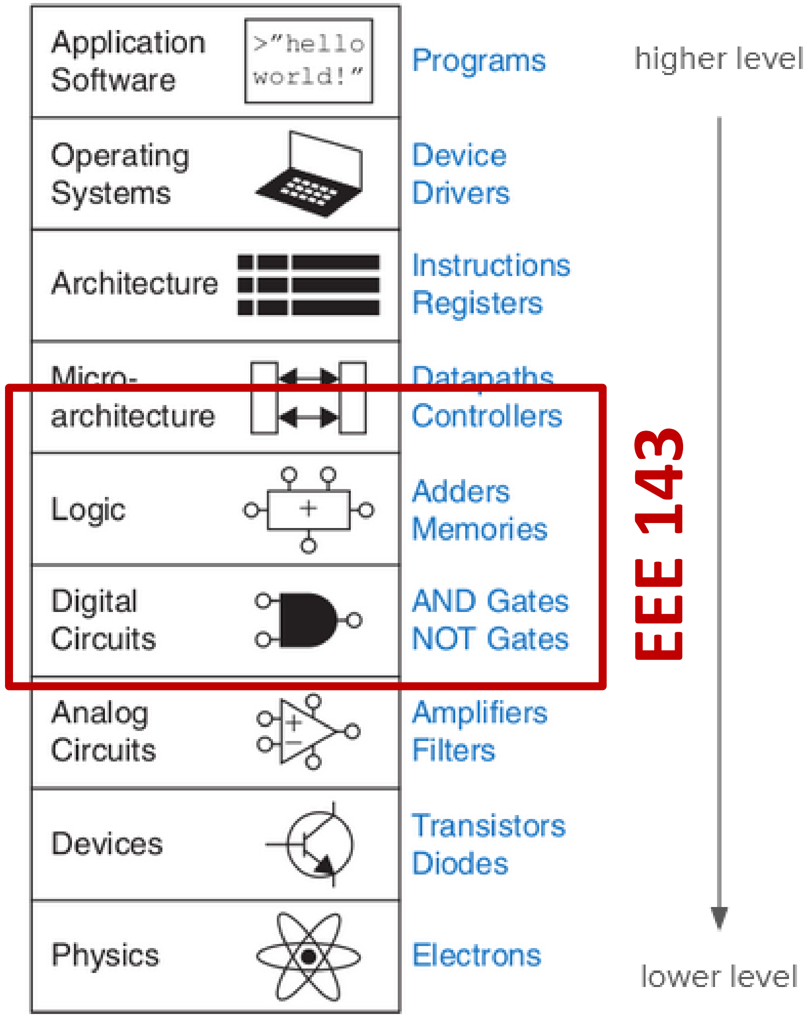


Image credit: *Digital Design and Computer Architecture* (2nd ed.) by Harris, H., Harris, S. (2013).

Up next for EEE 143

- Before we start manipulating digitized data, we need to know how to represent numbers in the digital domain

- **Number systems (for next week)**
  - Representing numbers in different bases (binary, hex, base *n*)
  - Representing fractional numbers
  - The IEEE-754 floating point representation